

5G: ARCHITECTURE AND KEY PRINCIPLES



5G



Outline

Overview

5G RAN

5G Core





Sathish



Let's get started!



5G: Architecture and Key Principles

A BRIEF HISTORY



OG: ERA OF GOING WIRELESS

1930s: Mobile Phone System (MPS)

Early
1900s:



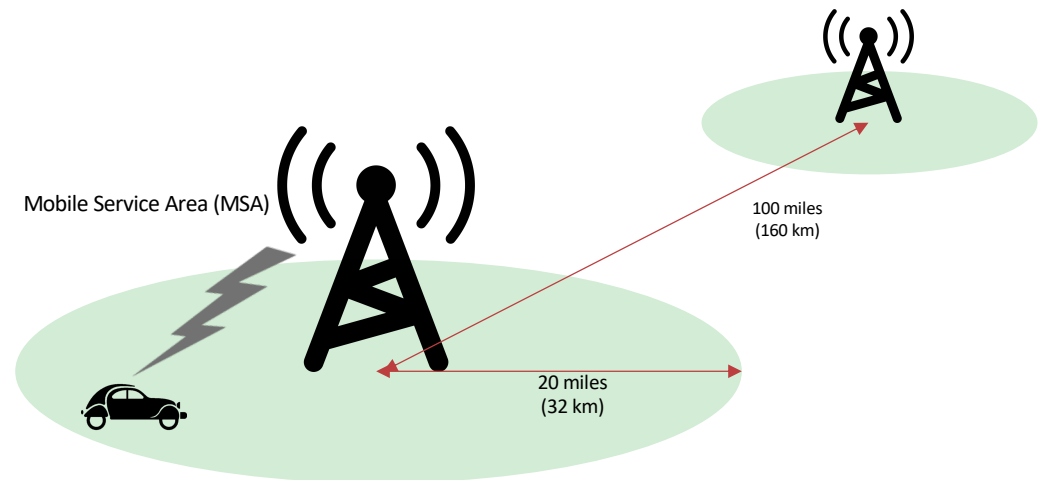
Push to talk service



Service Operator



A OG phone



- 10-30 channels from 36 MHz
- 1970s, 1980s: waiting list of 3 years

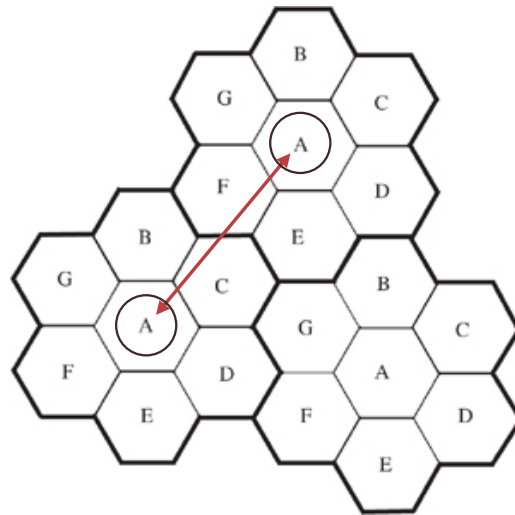
Key feature: Push-to-talk wireless phone



1G: ERA OF MOBILITY

1980s: Advanced Mobile Phone System (AMPS)

- 'Simultaneous talk' feature
- Analog
- FDMA
- 30kHz channels
- 666 Channels



Cellular communication



Issues: Poor voice quality, poor battery life, large size, poor handoff, poor security etc...

Key new feature: A 'mobile' phone + simultaneous talk



2G: ERA OF 'DIGITAL' COMMUNICATIONS

1990s: Global System for Mobile Communications (GSM)

- FDMA + TDMA
- SMS
- Packet switching based 2.5G (EDGE)



2G Phone

Key new feature: SMS



2.5G Phone

Key new feature: Email, web browsing



3G: ERA OF MOBILE INTERNET

2000s: UMTS/CDMA and
CDMA2000

- 5 MHz channels
- CDMA
- High speed data: from 384kbps to 14 Mbps



3G Phone

Key new feature: High-speed internet service, video calling, multimedia chat, etc



4G: ERA OF MOBILE 'BROADBAND'

2010s: EPS(LTE)

- 1.4 MHz to 20 MHz channels
- OFDMA downlink, and SC-FDMA in uplink
- MIMO
- Carrier aggregation

Milestone: more subscriptions than people!



4G Phone

Key new feature: high definition mobile television, mobile gaming, etc



5G: ERA OF _____?

Late 2010s: 5G (NR)

- Up to 400 MHz channels
- OFDM based air interface



5G Phone



5G-connected things

Key new feature: ?



THANK YOU

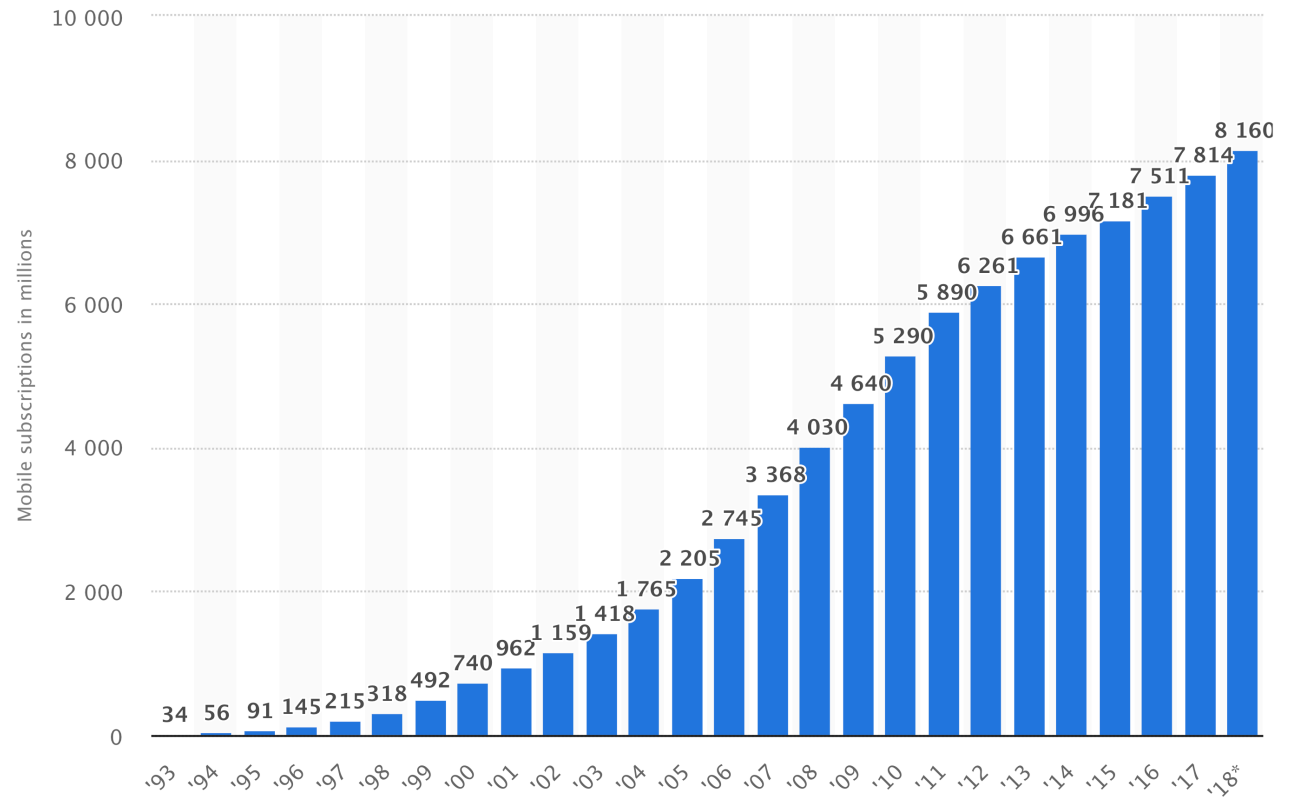


5G: Architecture and Key Principles

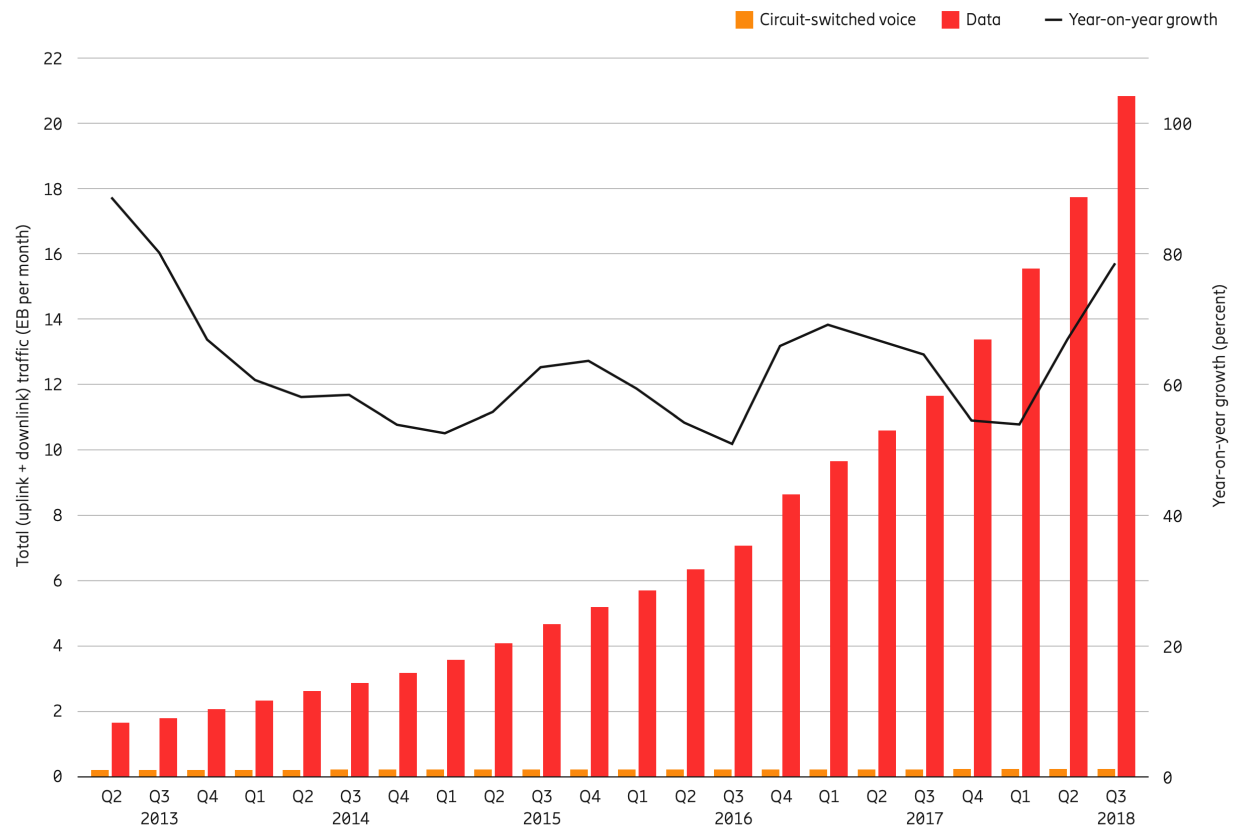
WHY 5G?



Subscription trends

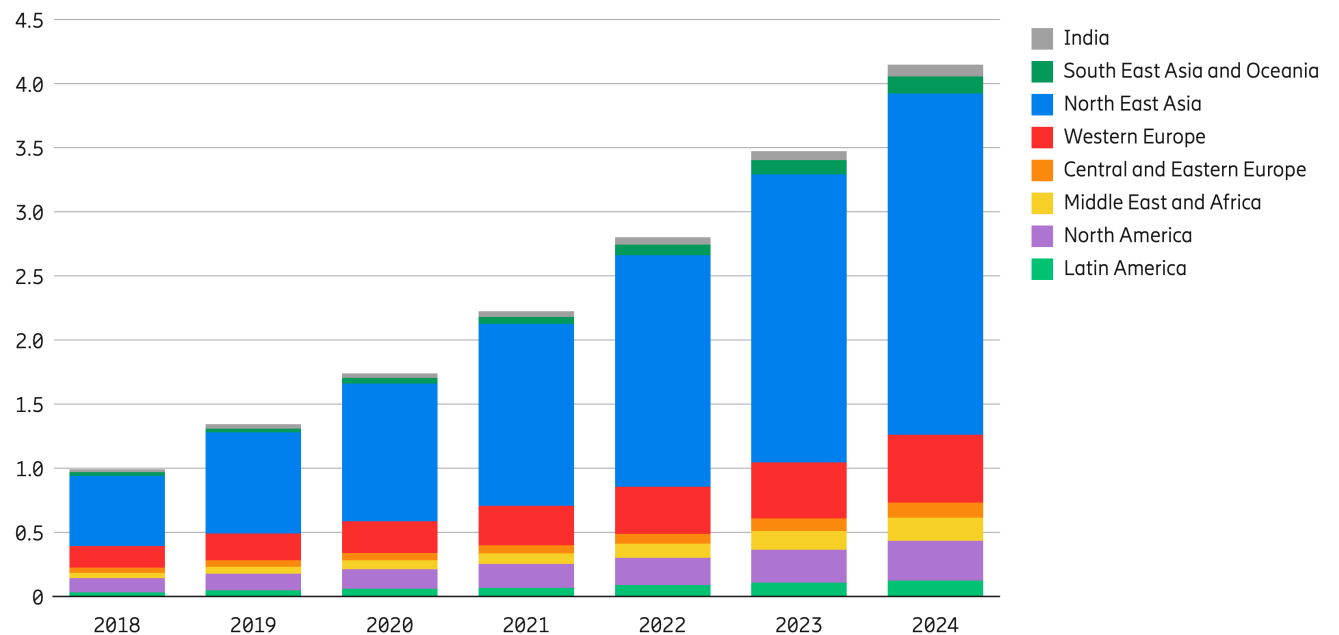


Data consumption trends



Internet of Things trends

Cellular IoT connections per region (billion)



METIS 2020

"The developed concept should support the following METIS overall technical goals (relative to 2013):

- 1000 times higher mobile data volume per area
- 10—100 times higher number of connected devices
- 10—100 times higher typical user data rate
- 10 times longer battery life for low power MMC, and
- 5 times reduced End-to-End latency"

"achieve these at a similar cost and energy consumption as today's (2013) networks"



5G PPP

The 5G Infrastructure Public-Private Partnership



WHAT IS 5G?



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5G:
Architecture
and Key
Principles

THE 5G USE CASES



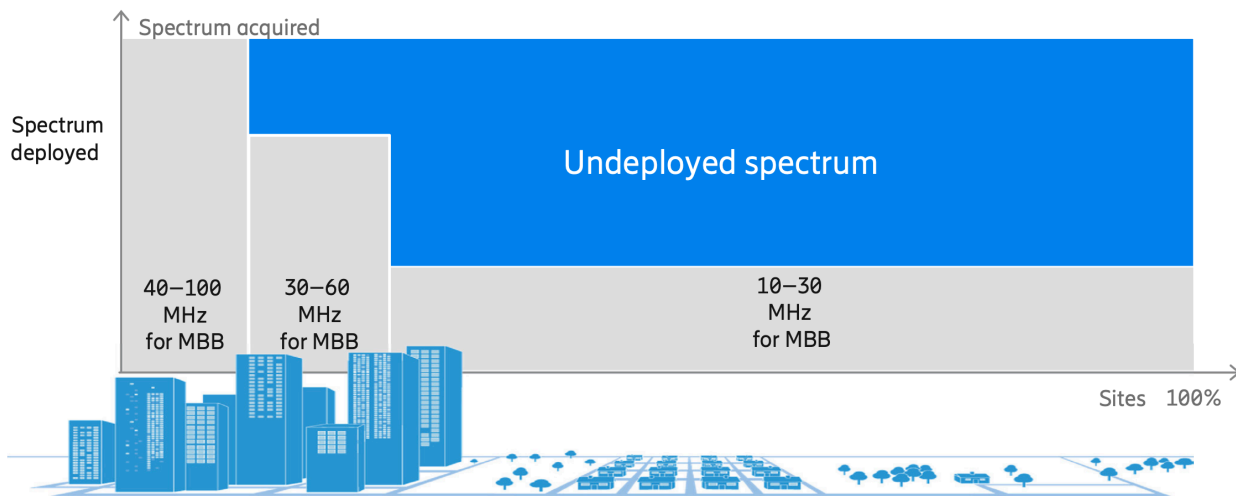


4G: Mobile Broadband

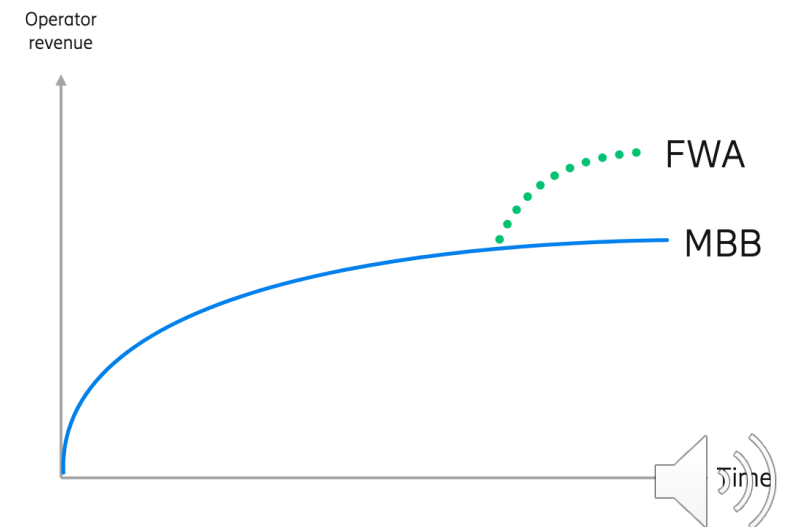
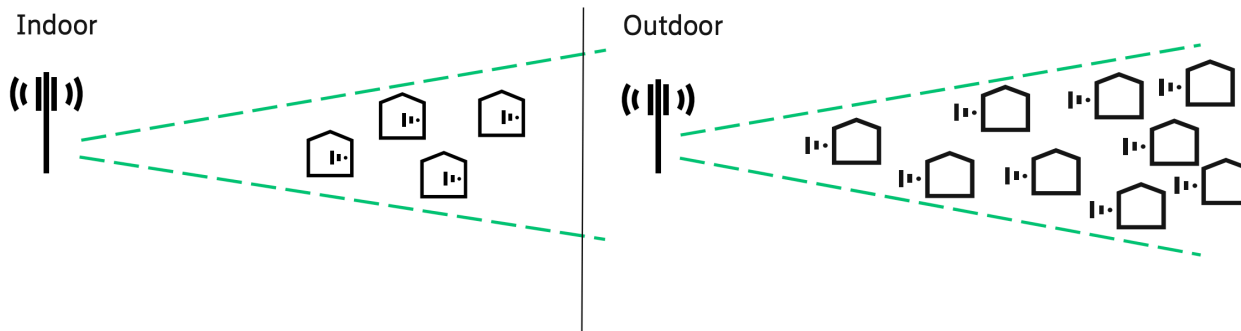


1) ENHANCED MOBILE BROADBAND





2) FIXED WIRELESS ACCESS (FWA)

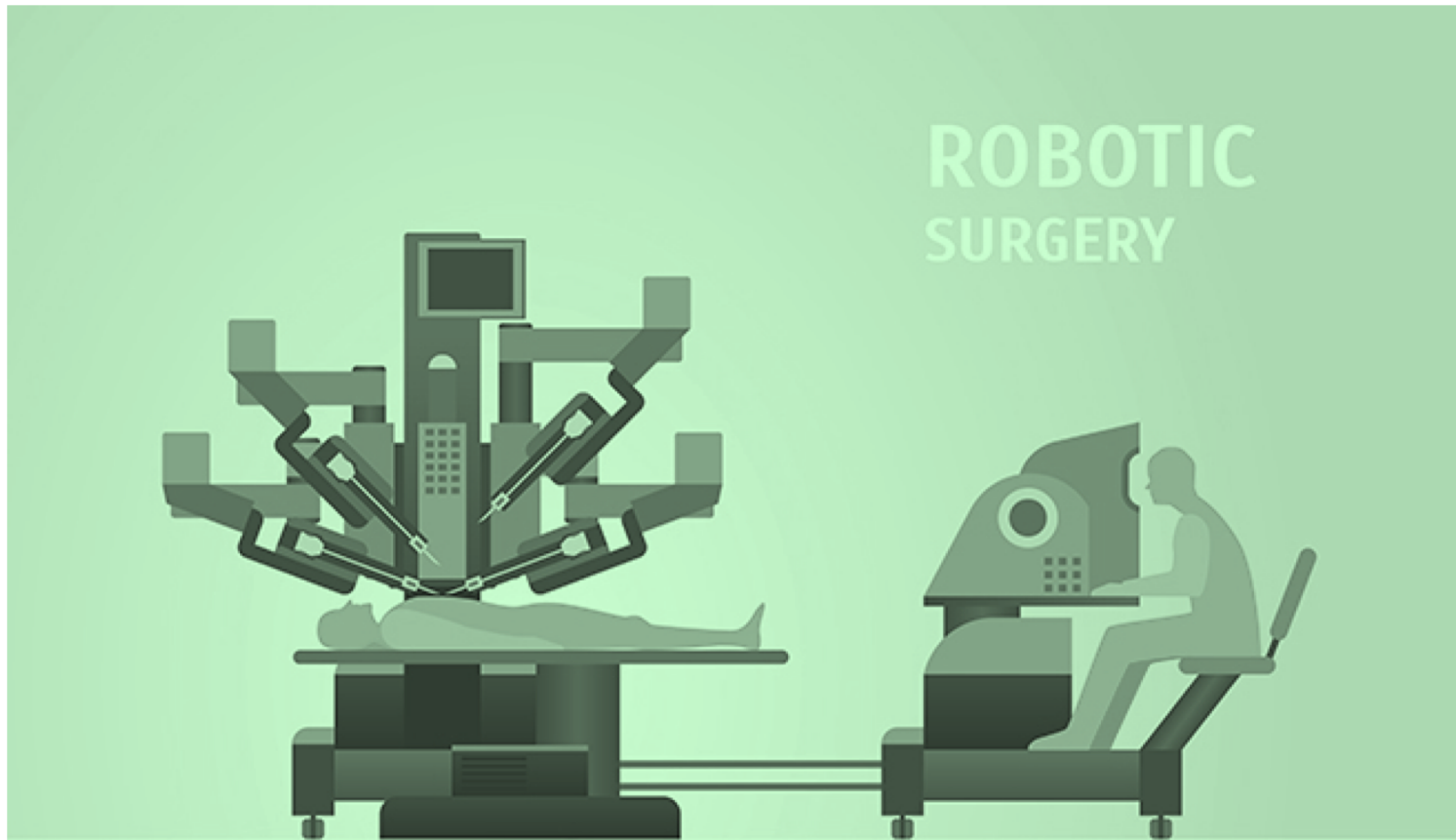


3) MASSIVE MACHINE TYPE COMMUNICATIONS (MMTC)

- Low data rate
- Low data volume
- Infrequent data
- Large number of devices
- Long battery life



4) ULTRA RELIABLE LOW LATENCY COMMUNICATION (URLLC)



- Ultra reliable
- Low Latency



Summary: Key 5G Use cases

- Enhanced mobile broadband
- Fixed Wireless Access
- Massive Machine Type Communications
- Ultra Reliable Low Latency Communications



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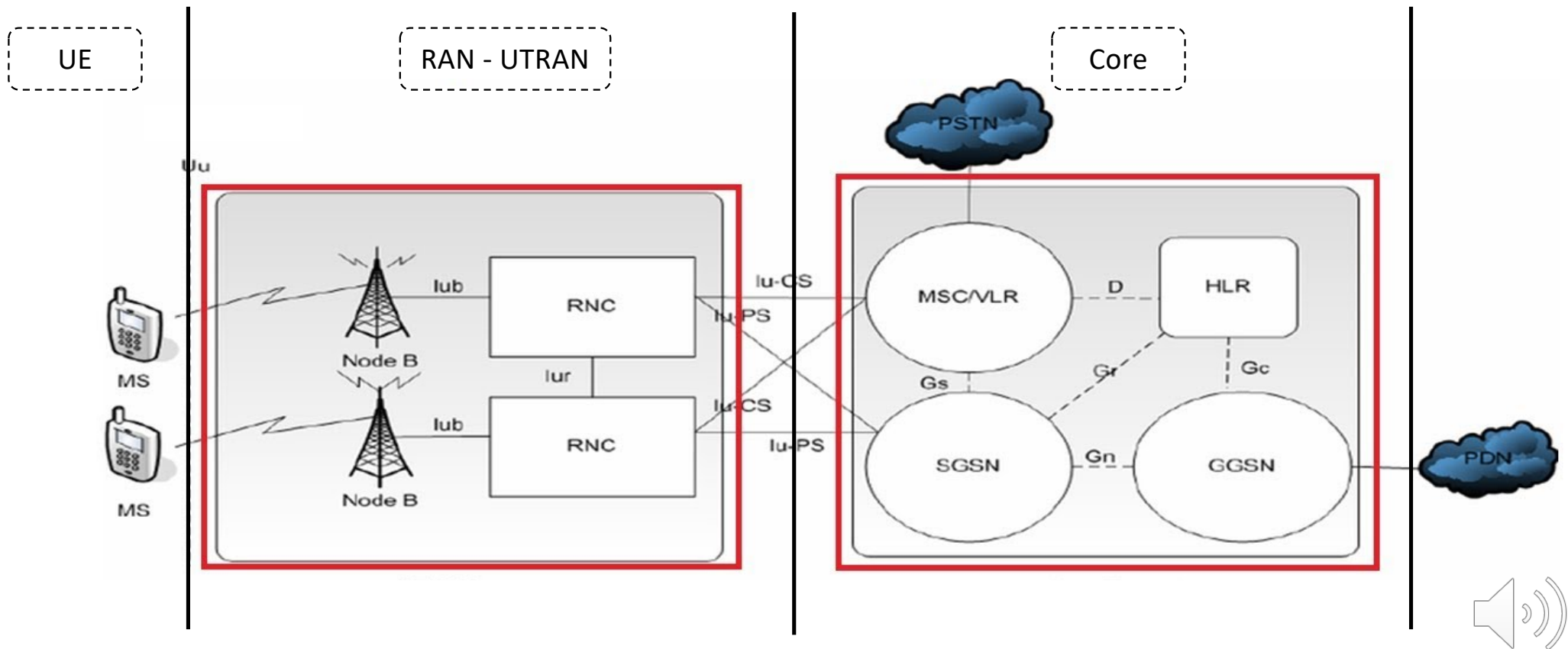


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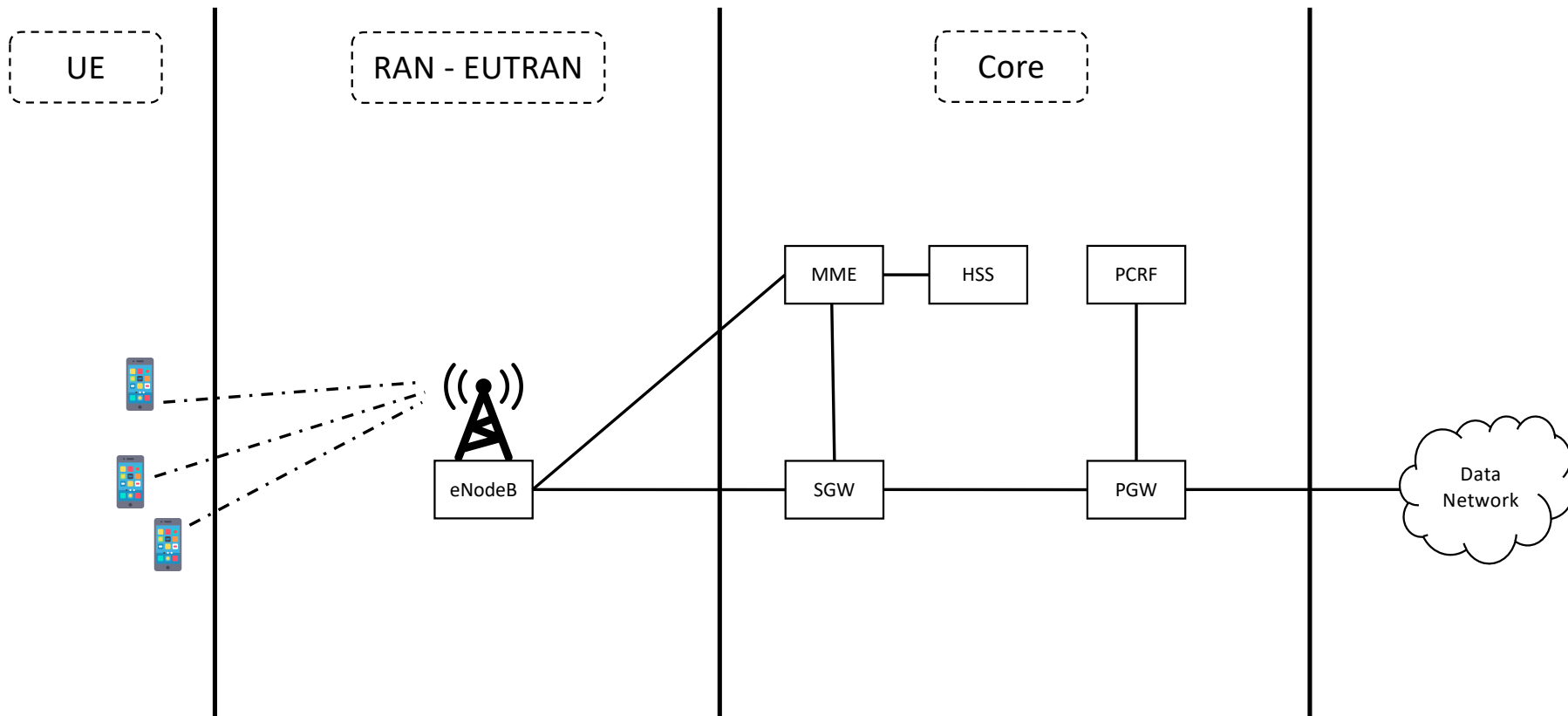
THE 5G SYSTEM



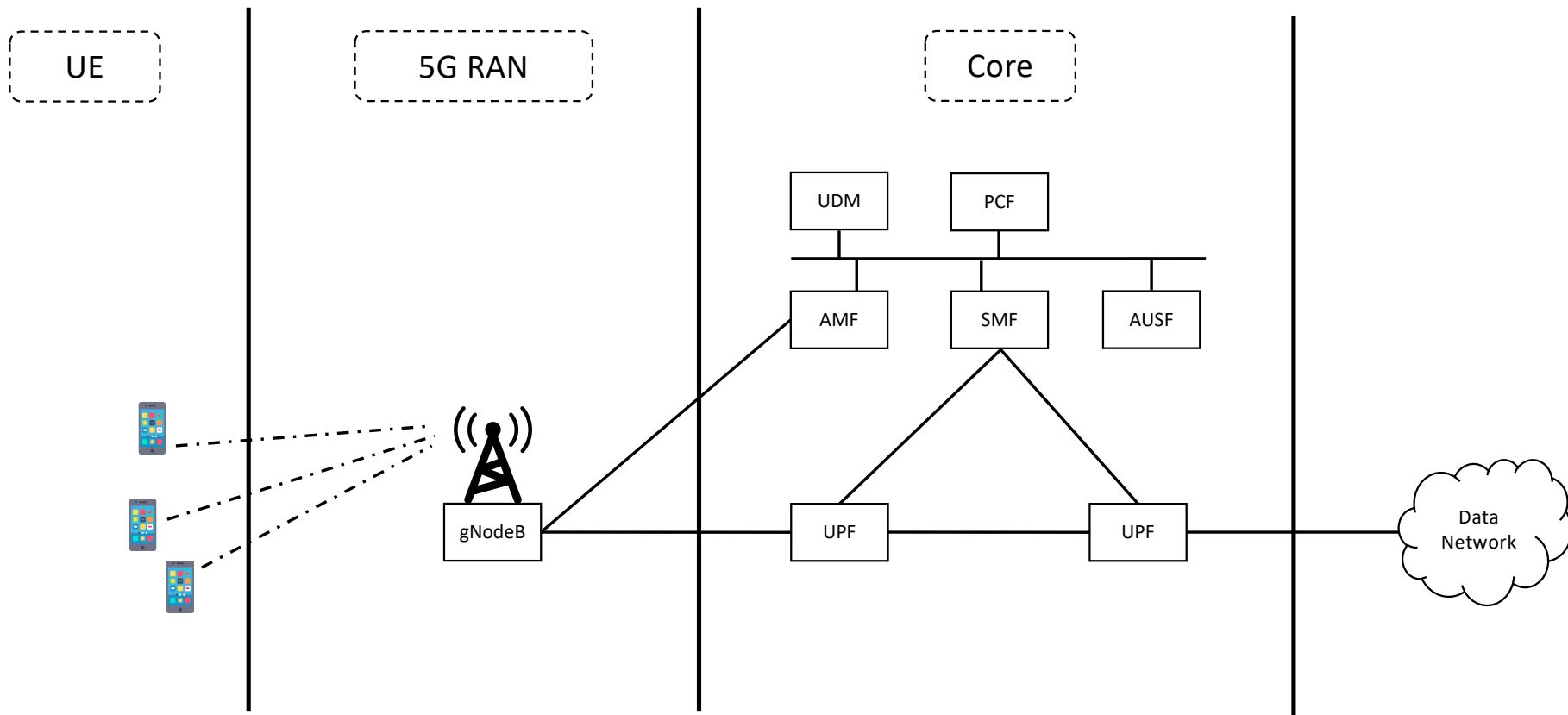
The 3G System



The 4G System



The 5G System



Control and User Plane Separation (CUPS)

Massive MTC:

- Very little data
- Large number of devices

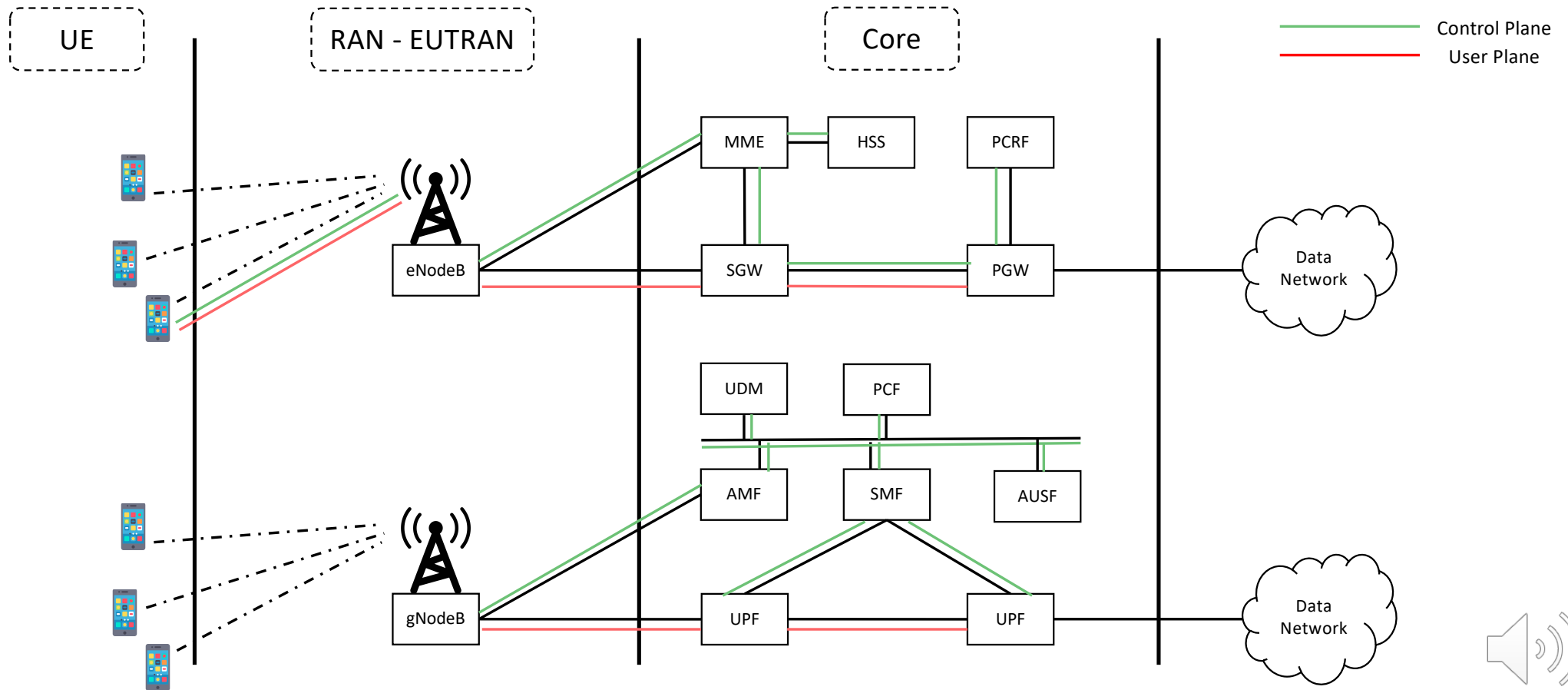


Mobile broadband:

- Lot of data
- Relatively fewer devices



Control and User Plane Separation (CUPS)



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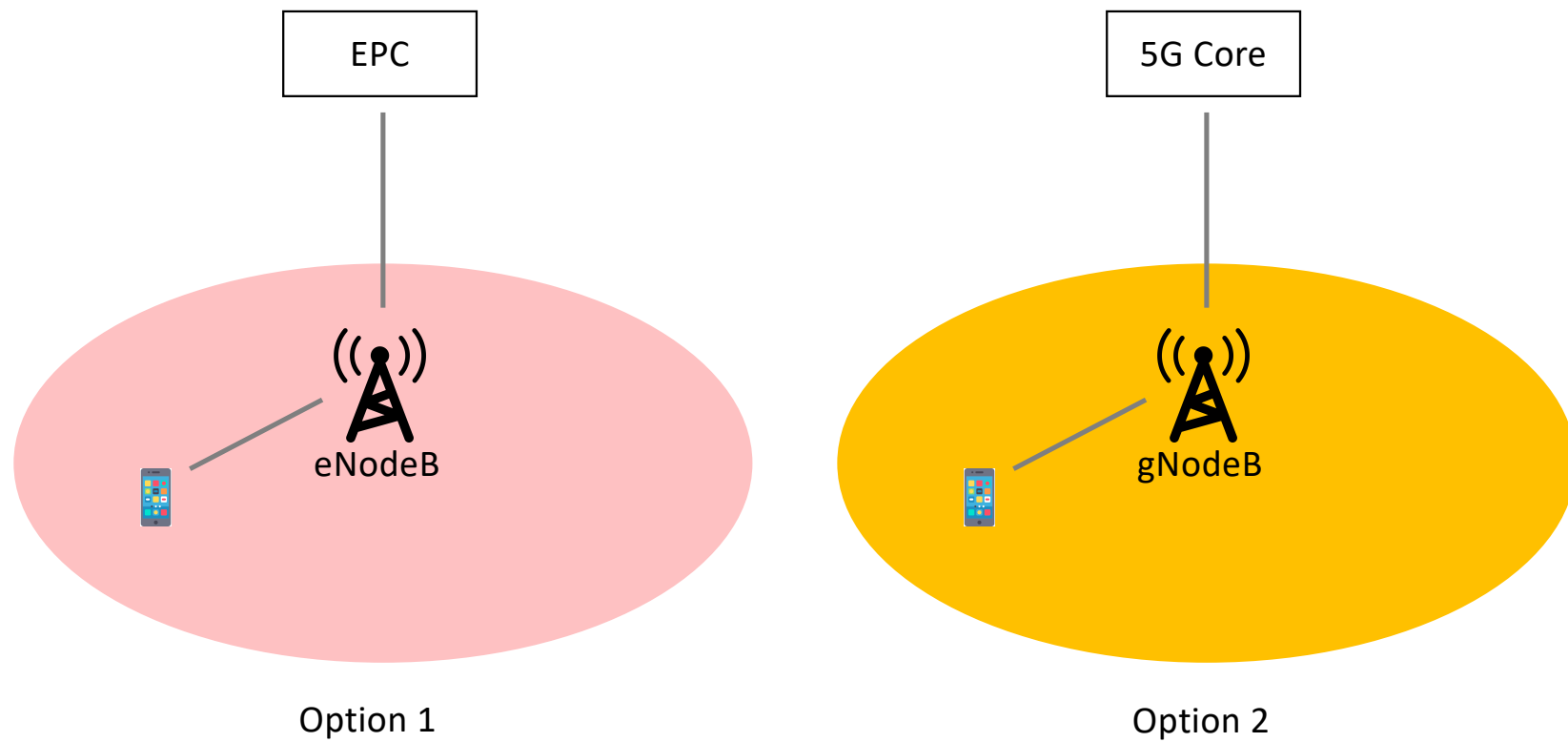


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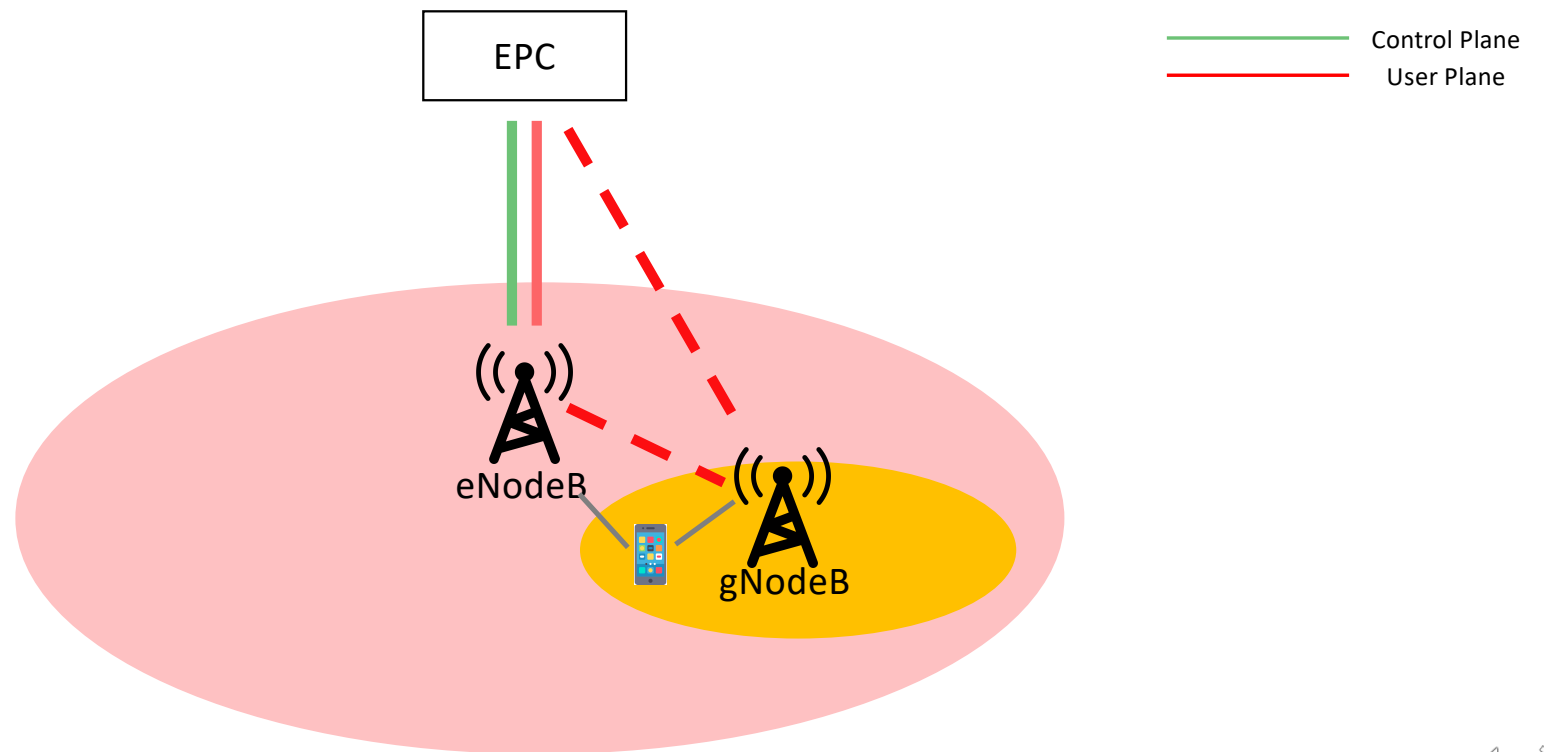
5G DEPLOYMENT OPTIONS



Option 1 and Option 2



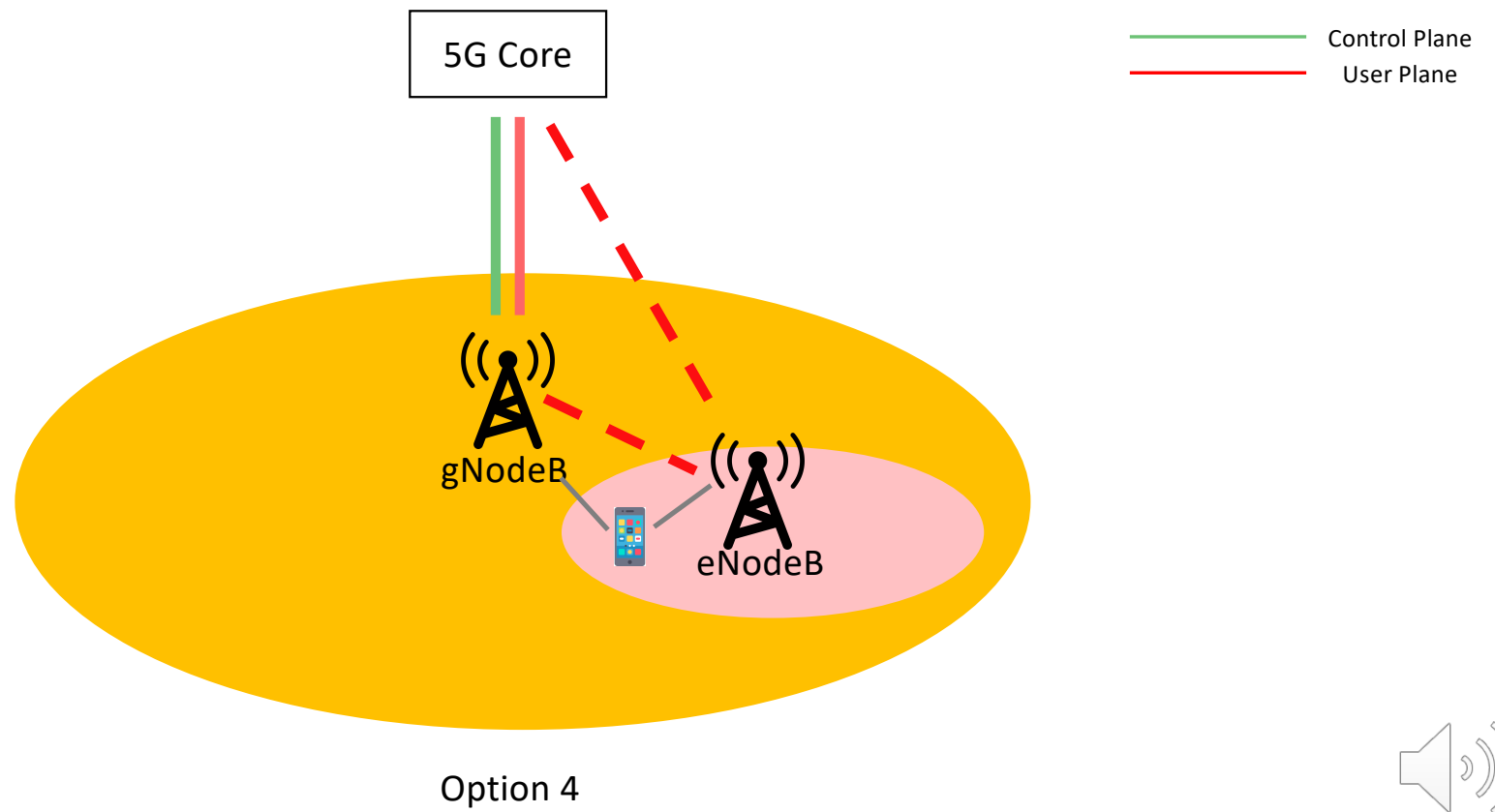
Option 3 – EN DC



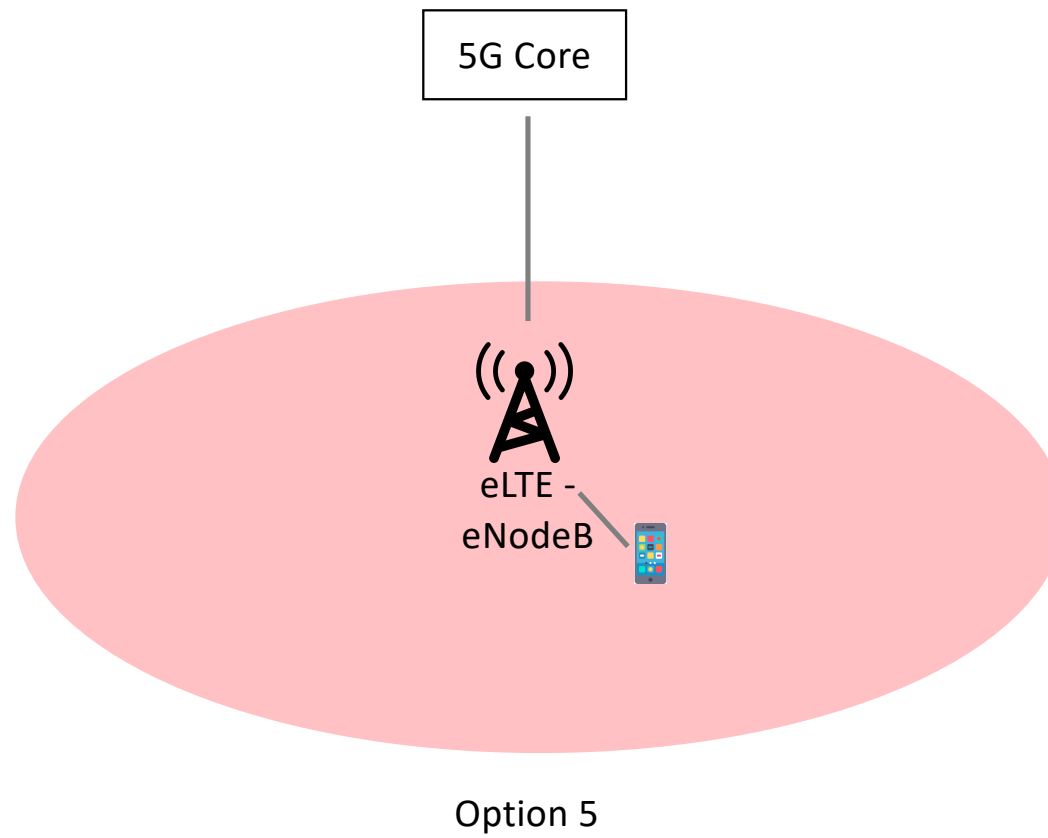
Option 3



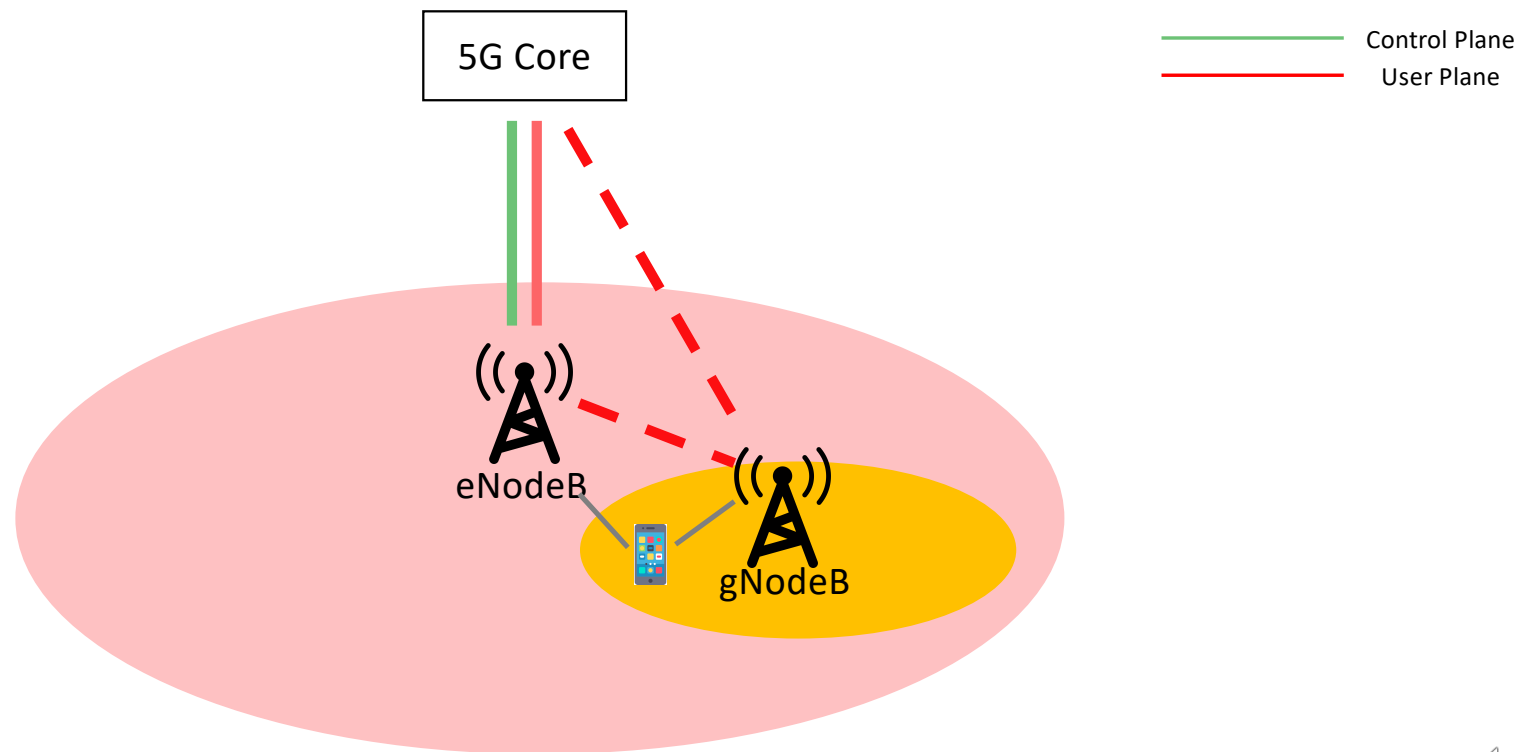
Option 4 – NE DC



Option 5 – eLTE



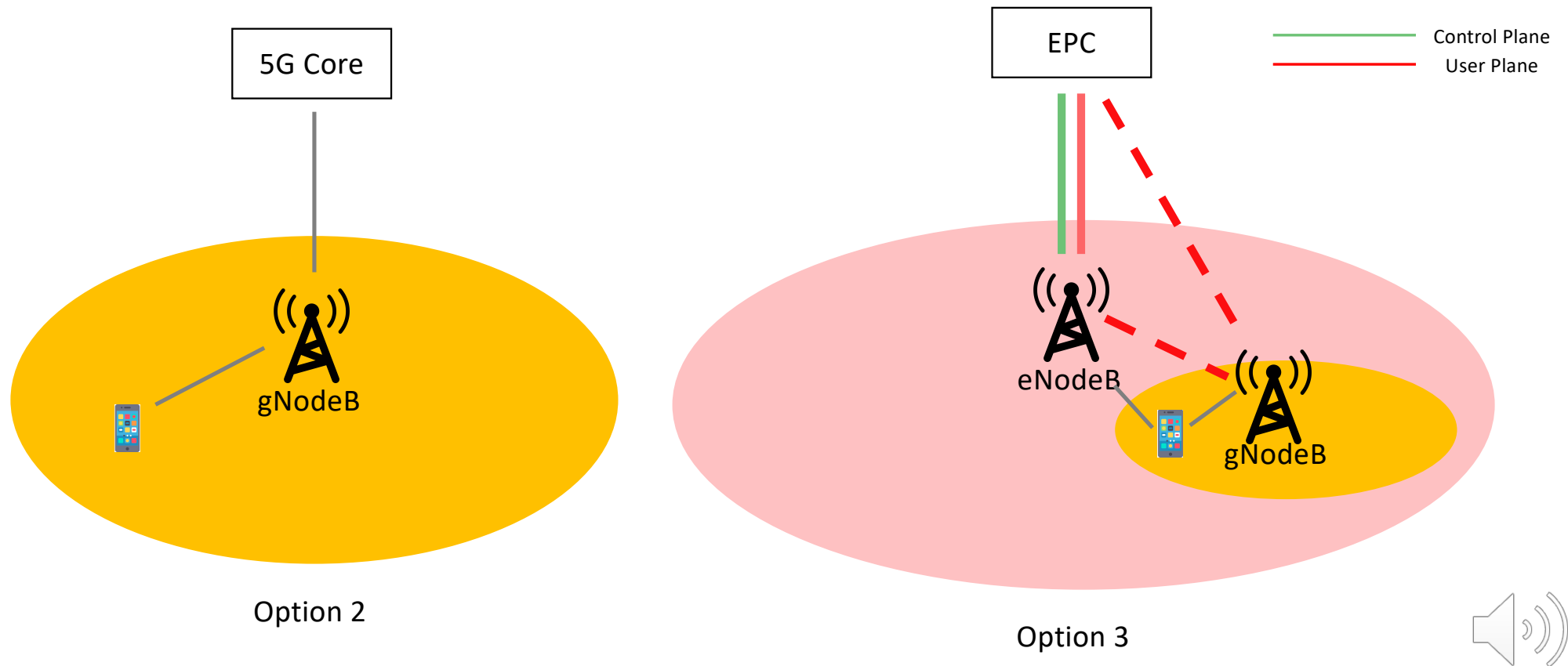
Option 7 – NGEN DC



Option 3



Non-standalone and Standalone



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